

NIKKO DUMRIQUE

☎ (604)-396-4817 | ✉ dumriquenikko@gmail.com | 🌐 nkoda | in nikkodumrique | 🏠 nkoda.github.io

SUMMARY

Backend engineer specializing in cloud infrastructure and production AI systems. Builds resilient, cost-efficient serverless platforms that make large-scale data (50TB+) accessible for analytics and ML, and ships LLM/RAG pipelines from retrieval benchmarking through deployment.

TECHNICAL SKILLS

- Programming:** Python, JavaScript, SQL, NoSQL (MongoDB)
- Cloud & Infrastructure:** AWS (Lambda, Lambda@Edge, S3, CloudWatch, API Gateway, Cognito), GCP (VMs, Networking), Serverless Architecture
- AI/ML:** RAG Pipelines, AWS Bedrock, Qdrant (Vector Search), Embeddings (Cohere Embed V4), Retrieval Evaluation (P@1, MRR), Model Deployment (Flask, Unicorn), LightGBM
- Data Engineering:** ETL Pipelines, Airflow, Redshift, S3 Data Lakes, Data Modeling, High-throughput Processing
- Systems:** Observability, Error Tracking & Auto-Recovery, Performance Optimization, Cost Optimization, System Design

EXPERIENCE

- Senior Software Developer (Data Infrastructure & Analytics)**
Responsive Ads

Jan 2024 – Present
Vancouver, BC

 - Architected and led development of a production RAG pipeline on AWS Bedrock and Qdrant, leveraging Cohere Embed V4 semantic retrieval across 400K+ ad creative embeddings to automate structured JSON layout generation; established retrieval benchmarks (P@1, MRR).
 - Managing a team executing Lambda backend modernization, implementing blue-green deployments, structured error framework, modern coding patterns, and enhanced CloudWatch observability to improve deployment safety and overall system reliability.
 - Built backend data infrastructure supporting a 55TB Redshift warehouse, 3TB S3 data lake, and 700GB MongoDB dataset for scalable analytics and ML/AI training; onboarded a UBC Masters of Data Science capstone team onto it by architecting isolated cloud environments (IAM, S3, MongoDB, GCP) to sandbox their work, building custom Python libraries for tooling access, and mentoring them on data normalization.
 - Identified cost and performance inefficiencies from billing and Zendesk analytics, then implemented automated management scripts (S3 lifecycle policies, GCP Compute Engine cron jobs, Lambda schedulers) that reduced storage footprint by 42% and improved system stability.
- Backend Software Developer (Data Infrastructure & Analytics)**
Responsive Ads

Sept 2023 – Jan 2024
Vancouver, BC

 - Led development of a platform-wide error tracking and auto-recovery system, reducing monthly engineering support tickets by 86% and improving system reliability and response time during critical failures.
 - Identified and executed the migration of analytics tracking from EC2 to Lambda@Edge, cutting monthly infrastructure costs by approximately \$7,500 (65% reduction) and improving response latency by 9% through serverless optimization.
 - Designed and built an in-house event-driven image optimization service using API Gateway, Cognito (P2P authentication), Lambda, and S3 to replace an unreliable third-party API and reduce monthly costs by 98% while improving system uptime.
- Software Developer (Full Stack)**
Emerging Media Labs

May 2020 – Aug 2021
Vancouver, BC

 - Developed automated ETL pipeline supporting multiple data formats (XML, JSON, CSV) with data validation and quality checks, reducing processing time by 82%
 - Conceptualized, coded, and documented a SQL database and accompanying Python scripts to handle large amounts of data for OLTP usage, resulting in a 33% improvement in first-byte latency.

PROJECTS

- Data Engineer: Financial Data Pipeline**
Airflow, LightGBM, Docker

 - Constructed financial data pipelines using Apache Airflow, Flask, Docker, and Python to predict trading volume from historical data, improving forecasting RMSE by 20% with parallel thread-pool ingestion that cut execution time by 47%.
 - Deployed a LightGBM regression model through a Flask API using Unicorn and CircleCI, enabling real-time predictions and reducing deployment time by 52%.

EDUCATION

- University of British Columbia**
B.Sc in Physics, Minor in Data Science

Vancouver, BC
Sept 2017 – May 2022